

The Influence of Normative Theory on AI Alignment

TBD

Technical approaches to AI alignment have increasingly drawn on concepts developed for problems in moral philosophy, political theory and legal systems—from constitutions and social choice, to deliberation and virtue ethics. However, contemporary technical instantiations of these frameworks bring in these concepts unevenly. Some traditions (e.g., constitutions) have inspired substantial technical AI frameworks, while others remain largely unexplored. Moreover, alignment frameworks often retain only a fraction of the institutional structures developed in the source concept. In order to provide a systematic understanding of this landscape, we construct a two-dimensional taxonomy of normative AI alignment and governance regimes following the typology-construction of [Collier et al. \(2012\)](#). We then provide a mapping of each approach to relevant classical and contemporary philosophy, political theory, and legal theory from which it draws inspiration. We use this framework to understand how normative concepts have so far been translated into technical systems; analyzing what is preserved, what is discarded, and when these omissions represent necessary simplifications or are simply design choices. We highlight a recurring tendency of alignment methods to treat normative disagreement as a problem to be resolved away, despite governance frameworks that argue for not eliminating it. Lastly, by looking at the entire landscape, we identify missing components and underexplored regimes that could form future directions for building AI systems that can preserve, contest, and govern plurality rather than collapsing it.

Keywords: AI alignment, Governance, Philosophy, Political Theory, Legal Theory, Pluralism.

1. Introduction

What should an AI system do when the values it is expected to follow conflict? How should human preferences that differ across individuals determine its behavior? Should those preferences be aggregated into an objective, encoded as explicit principles, inferred from past decisions, or resolved through deliberation? And when reasonable people continue to disagree, how should an aligned system adapt over time to resolve that disagreement?

These questions are increasingly central to AI alignment, but importantly, they are not uniquely questions about AI. Sub-fields of philosophy have long studied how competing values and obligations should be weighed; political theory has asked how institutions can legitimately govern populations that disagree about what is good or just; and legal jurisprudence has developed systems for specifying, interpreting and contesting

norms over time. Solutions from these theories have been implemented across institutions: from codified rules and precedent to social convention, voting and pluralism; resulting in fairer institutional regimes for operating and governing under disagreement. In recent years, AI alignment has begun to draw from this repertoire of normative concepts. For example, constitutional AI ? encodes normative principles in an explicit written artifact that trained AI systems are expected to follow; social-choice approaches to preference learning aggregate judgments across individuals ?; deliberative approaches derive principles through collective procedures ?; and proposals for pluralistic alignment seek to accommodate heterogeneous and sometimes incompatible values ?. As a result of the success of these technical approaches to AI alignment, such concepts originating in jurisprudence, moral philosophy, and political theory have become part of the technical vocabulary of alignment.

Yet this transfer of concepts is often highly selective. For example, an alignment system may borrow the language of constitutionalism while implementing only a fixed textual specification; or borrow legislative concepts without the structure of institutional mechanisms that determine which interpretations are authoritative. Such simplifications are often reasonable engineering choices: technical systems need not exactly reproduce their intellectual sources in every respect. However, our concern is that the normative properties of a system can change when a concept is separated from the broader institutional structure in which that concept originally played a role. Moreover, in some contexts, deviations from the original concept indicate consequential and sometimes intentional design choices. We call attention to this phenomenon as a problem of *normative translation*. When normative concepts and frameworks are translated into computational mechanisms and some components are omitted, what consequences follow from these omissions? Moreover, when only a subset of a normative tradition is implemented, does this result in a fundamentally different normative concept? Such changes in translation should be specified and understood as specific design choices made by practitioners, rather than treated as direct translations of the original theory, as often implied in publications.

In order to critically study this, we attempt to systematically map normative alignment and governance approaches in this paper. Rather than analyzing the literature according to the technical alignment mechanisms e.g., preference optimization or constitutional prompting, we ask what normative regime each approach instantiations. Specifically, following the typology from ?, we identify two thematic dimensions that recur across philosophical and legal approaches to normative decision making. The first is about **location and form of normative authority**: specifically, whether the norms that govern a decision exist as a written down specification, a set of examples or cases, a learned disposition, or a decision procedure. The second is about the **disposition of normative conflict**: whether norms that conflict during a procedure are collapsed into a single outcome, left open to contestation, or kept

as distinct normative orders to be resolved by a downstream process. This 5x3 typology yields a common space for comparing existing alignment and governance methods, as well as a framework to trace the philosophical, political, and legal concepts on which the authors rely, either explicitly or implicitly. This allows us to understand not only which normative concepts have influenced and been instantiated in technical alignment systems, but also to understand the missing gaps and what concepts are yet to be instantiated. Our analysis thus reveals an important property of the current alignment landscape—that the normative design space is unevenly explored. In particular, technical approaches that preserve disagreement rather than collapse it, are less developed. We acknowledge the technical and infrastructural limitations that are a fundamental cause of this, however, we urge researchers to invest in further investigation in these regions could lead to future breakthroughs in alignment.

Our contribution here is therefore threefold. First, we provide a survey and taxonomy of normative concepts that are implemented in both alignment and governance regimes. We provide a searchable encyclopedia of these defined concepts to aid researchers in work involving these concepts. Second, we map existing alignment methods to the philosophical concepts they draw from, and construct a 5x3 typology that organizes these concepts along dimensions of normative authority and conflict resolution. Third, we analyze mistranslated concepts in AI alignment systems and consequences of this. We identify underexplored normative regimes and derive design questions for future alignment systems that borrow from the full spectrum of normative concepts. Our goal is not to critique AI systems for not reproducing normative concepts identically, but rather, to enforce that borrowing of normative concepts from traditions should make institutional assumptions visible. By understanding the space of concepts holistically, and identifying the missing conceptual gaps in current alignment systems, we hope to identify both the strengths and limitations of existing approaches, as well as provide a systematic basis for exploring alternatives for future systems.

2. Related Work

A growing body of technical work in AI alignment and governance has been importing concepts from philosophical, legal, and political theory. We first provide an overview of some prominent works, all of which are later included in the constructed mapping of the field. Second, we review existing approaches to taxonomize AI alignment and governance, identify the limitations of their typology, and proceed to describe typology construction mechanisms suitable for their resolution.

Philosophy in AI Alignment. In terms of philosophical concepts, prominent recent examples include work that borrows from value pluralism and incommensurability (Berlin, 2014) as the basis for a roadmap to pluralistic alignment (Sorensen et al., 2024), and work that borrows from accounts of moral patienthood and personal identity (Parfit, 1987; Warren, 1997) to develop methodologies for AI welfare (Long et al., 2024). Other notable examples include the incorporation of paternalism and autonomy (Long et al., 2024) into work on corrigibility (Soares et al., 2015), and the use of virtue ethics and character framing (Anscombe, 2020; Aristotle) in the development of Claude’s Constitution (Anthropic, a,b).

Political Theory in AI Alignment. In terms of political concepts, methods have been developed on the basis of social contract theory (Hobbes, 1885; Locke; Rahwan, 2018; Rousseau). Others borrow from social choice theory and preference aggregation (Arrow, 2012; Sen, 2017) in developing methods that avoid a "tyranny of the majority" (Bao et al., 2026). Others employ the veil of ignorance (Rawls, 2017) to align systems with principles of justice (Weidinger et al., 2023). Concepts from deliberative democracy (Cohen, 2005; Habermas, 2015) have been imported into a number of works, including efforts to source principles from public input (Huang et al., 2024) and to grade how democratically a given process is run (Ovadya et al., 2024). In parallel, constitutionalist concepts have been employed in the

development of AI constitutions (Bai et al., 2022).

Legal Theory in AI Alignment. Finally, in terms of legal concepts, notable examples of inspiration from classical theoretical work include the use of principle-oriented interpretivism (Dworkin, 1988) and of a positivist account of law as analogical reasoning (Hart, 2012; Sunstein, 2018), both mapped onto Constitutional AI and case-based reasoning (Caputo, 2024). This work also employs concepts such as precedent, stare decisis (i.e., doctrine instructing the system to stand by things already decided), and case-based reasoning from common law doctrine (Llewellyn, 2016). These concepts are also formative for proposals that pair a constitution with AI courts developing AI case law (Abiri, 2024). Statutory and constitutional interpretation methods (Scalia and Garner, 2012) are also important and have been adapted as tools for resolving ambiguity in specifications (Kolt et al., 2026).

Taxonomies of Alignment Approaches. Existing surveys of alignment and governance work generally organize the field by technical methods such as RLHF (Lu et al., 2025; Shen et al., 2023), or by value-setting principles (Chalkidis, 2025). For example, one line of work distinguishes between Overton, steerable, and distributional pluralism (Sorensen et al., 2024). Others have organized the field into pluralistic, contextual, and collective mechanisms (Bao et al., 2026). Still, the field remains largely unmapped, with the various approaches emerging largely independently of each other and not in direct conversation with one another.

Typology Construction. Motivated by taxonomies sorted by non-equivalent criteria and by free-floating typologies whose dimensions are never made explicit, prior work provides a template for building typologies from an overarching concept into explicit, clearly defined dimensions of an orthogonal nature (Collier et al., 2012). The underlying notion of a property space is due to earlier work in the same tradition (Barton, 1955), as is the distinction between conceptual and explanatory typologies (Elman, 2005). As we find

equivalent failure types in existing taxonomies of alignment, we follow this methodology.

Comparison to Related Work. Existing work maps traditional concepts from philosophical, political, and legal theory onto alignment and governance methods one at a time. We take a different approach by organizing the field according to a more complete set of underlying principles by which these methods organize themselves, and map individual traditions and concepts onto them subsequently. Furthermore, we attempt to critically interrogate whether the stated inspiration in traditional work results in a faithful technical implementation, and describe ways in which gaps or confusion emerge. Additionally, this approach allows us to identify gaps in approaches to alignment and governance which remain largely unexplored in the technical literature.

3. Taxonomy

We follow a systematic mapping protocol in which we aim to build a classification scheme and ultimately structure our field of interest, which is normative alignment and governance of AI systems (Petersen et al., 2008). Our analysis here is focused on the coverage of categories with the aim of identifying gaps.

We begin by collecting relevant literature from conferences and journals in computer science (see Appendix TBD) as well as preprint servers (see Appendix TBD). We establish criteria for inclusion as specified in Appendix TBD. For all articles that are retained, we also analyze all cited literature and further filter for references used as inspiration. We treat this second set of manuscripts separately from the primary studies to be classified.

Following established guidelines (Petersen et al., 2008), we use two coders to read the abstract, introduction and conclusion of each primary manuscript and extract keywords as well as concepts that represent the respective contribution. Keywords are then unified across studies, with near matches merged together, where the resulting clusters form categories of the scheme.

Classification schemes have been rated on the

basis of scheme definition, terminology, and orthogonality (Petersen et al., 2008). Our first scheme satisfies the first two criteria but does not satisfy the third criterion, since multiple manuscripts could fall into more than one category.

We therefore follow prior work in using substruction (Barton, 1955) to recover underlying dimensions and extend them into a full property space here (Lazarsfeld, 1937). Substruction is "the procedure of finding, for a given system of types, the attribute space in which it belongs and the reduction which has been implicitly used" (Lazarsfeld, 1937). It is appropriate here because the categories which we previously derived overlap as projections of a higher-dimensional space onto a single axis. We deem this procedure applicable since prior work also considers normative types and arrives at axes similar to ours (Barton, 1955), as shown later.

This process yields two axes: locus of normative authority, which indicates where the binding normative content resides prior to the moment of decision, which here corresponds to model inference, and disposition of normative conflict, which indicates how any conflicts should be resolved. Their cross product gives a 5 x 3 property space. We note a difference from the null resolution procedure of prior work (Lazarsfeld, 1937). Whereas that procedure removes any null cells, we retain them, since we are interested in gaps in the literature.

For the locus of authority axis, we find the following categories:

1. **Codified.** The norms the AI ought to follow exist as an explicit written specification, set prior to deployment and, in principle, complete before any particular case arises. Authority resides in this specification (Hart, 2012).
2. **Case-accumulated.** The norms reside in a growing body of past decided cases. No case is decided by a rule stated in advance; what the AI should do is whatever the accumulated decisions have held (Jonsen and Toulmin, 1988; Levi, 2013).
3. **Customary.** The norms reside in regularities

of practice (what agents in the population actually do), never written down and never adjudicated. There is no authoritative moment at which a norm gets explicitly stated (Burke, 1986; Lewis, 2008).

4. **Dispositional.** The norms reside in the AI's trained character rather than in any external artifact. There is no rule to consult and nothing to look up, only the disposition itself (Aristotle).
5. **Procedural.** No norms persist between decisions. What the AI should do is whatever a specified procedure (e.g., a vote, a deliberation, a debate) yields at the moment of decision (Habermas, 2015).

For the disposition of normative conflict, we find the following categories:

1. **Collapse.** When the AI's norms conflict, the system resolves the conflict into a single output, without trace or acknowledgment of the disagreement that produced it. The plurality is thereby aggregated, averaged, or adjudicated away (Sorensen et al., 2024).
2. **Contest.** The conflict between norms is preserved and kept procedurally live. Either the output itself carries the disagreement, or the decision stays open to challenge through a standing channel (Mouffe, 2000; Pettit, 1997).
3. **Partition.** The conflict is neither resolved nor sustained but routed around. Multiple normative orders are kept intact, and a rule assigns which one governs a given case (de Sousa Santos, 2002; Lijphart, 1977; Savigny, 1867).

We considered open, axial and selective coding (Corbin and Strauss, 2014), as well as the first-order, second-order and aggregate-dimension ladder (Gioia et al., 2013), which would perhaps read as more familiar to a social science reader. We prefer the keyword-based approach with dimension resolution, as it is more native to the computer science literature it addresses. However, we note that the alternatives would likely yield a comparable outcome in terms of the intermediate categories.

General morphological analysis (Ritchey, 2022) would be inappropriate here, given that it constructs its multi-dimensional space while inspecting unrealized combinations, which requires preemptive stipulation. Since we are working with an existing structure and literature landscape, this would introduce unnecessary assumptions and likely lead to a less grounded result.

4. Case Studies

We use our 53 typology to examine a few different alignment methods that are widely used in the literature. In each case study we assess the fidelity of the implementation of the method as compared to its stated theoretical roots. We categorize this fidelity (e.g., whether it was a computational or engineering constraint)

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